

Assignment 8

$$f(x) = \begin{cases} \frac{1}{x} & \text{if } x \in (0,1] \\ 0 & \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} \frac{1}{x} & \text{if } x \in (1,\infty) \\ 0 & \text{otherwise} \end{cases}$$

Determine the sets

$$P_1 = \{p \in [0, \infty]: f \in \mathcal{L}^p(\lambda)\}$$

$$P_2 = \{p \in [0, \infty]: g \in \mathcal{L}^p(\lambda)\}$$

We start with P_1

We start by checking 0 and ∞

" ∞ "

We check the definition we've been given

Can we find $R > 0: |f| \leq R$

The answer is no

$$\text{let } R > 0$$

$$\text{pick } 0 < d < \min\left\{\frac{1}{R}, 1\right\}$$

Then $f(d) = \frac{1}{d} > R$

This is also true for any $b \in (0, d)$

Meaning that $\{|f| > R\} \cap (0, d) = (0, d) \neq \emptyset$

Thus not a null set and

$$\neg(|f| \leq R \text{ a.e.})$$

Thus $\infty \notin P_1$

"0"

Again we check the definition

We define the set

$$I_t = \{x \in (0,1]: |f|(x) \geq t\}$$

Then its enough to show that $\lambda(I_t) \rightarrow 0$ for $t \rightarrow \infty$

$$\frac{1}{x} \geq t \Leftrightarrow x \leq \frac{1}{t}$$

$$I_t = \left(0, \min\left\{1, \frac{1}{t}\right\}\right] \subseteq \left(0, \frac{1}{t}\right]$$

when of course we can just drop the minimum when t grows

$$\lambda(I_t) \leq \lambda\left(\left(0, \frac{1}{t}\right]\right) = \frac{1}{t} \rightarrow 0 \text{ for } t \rightarrow \infty$$

Thus

$$\lambda(I_t) \rightarrow 0 \text{ for } t \rightarrow \infty$$

And we have $0 \in P_1$

Now for $p \in (0, \infty)$

We need to start calculating some integrals

We take the definition

$$f \in \mathcal{L}^p(\lambda) \Leftrightarrow \int_{\mathbb{R}} |f|^p d\lambda < \infty$$

So we start by thinking of the function we are integrating

$$|f|^p(x) = \begin{cases} \frac{1}{x^p} & \text{if } x \in (0,1] \\ 0 & \text{otherwise} \end{cases}$$

This is continuous on the half open interval and 0 otherwise so we can simply split it up and calculate lebesgue integral as the limit of riemann integrals

$$\int_{\mathbb{R}} |f|^p d\lambda = \int_{\mathbb{R}} \mathbb{I}_{(0,1]} \frac{1}{x^p} \lambda(dx) = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} \mathbb{I}_{[\frac{1}{n},1]} \frac{1}{x^p} \lambda(dx) = \lim_{n \rightarrow \infty} R \int_{\frac{1}{n}}^1 \frac{1}{x^p} dx$$

For $p \neq 1$

$$R \int_{\frac{1}{n}}^1 \frac{1}{x^p} dx = \left[\frac{x^{1-p}}{1-p} \right]_{\frac{1}{n}}^1 = \frac{1}{1-p} \left(1 - \left(\frac{1}{n} \right)^{1-p} \right)$$

For $p < 1$

$$\frac{1}{1-p} \left(1 - \left(\frac{1}{n} \right)^{1-p} \right) \rightarrow \frac{1}{1-p} \text{ for } n \rightarrow \infty$$

Since $\left(\frac{1}{n} \right)^{1-p} \rightarrow 0$

$$\Rightarrow (0,1) \subseteq P_1$$

For $p > 1$

$$\frac{1}{1-p} \left(1 - \left(\frac{1}{n} \right)^{1-p} \right) \rightarrow \infty$$

Since $\left(\frac{1}{n} \right)^{1-p} \rightarrow \infty$ and $\frac{1}{1-p} < 0$

For $p = 1$

$$R \int_{\frac{1}{n}}^1 \frac{1}{x} dx = [\log(x)]_{\frac{1}{n}}^1 = -\log\left(\frac{1}{n}\right) = \log(n) \rightarrow \infty \text{ for } n \rightarrow \infty$$

Thus any number in $[1, \infty)$ are not in P_1 and thus we have $P_1 = [0,1)$

Now to P_2

Again we should check $0, \infty$

" ∞ "

Yes its

since

$$|g|(x) \leq 1 \quad \forall x \in \mathbb{R} \Rightarrow |g| \leq 1 \quad \lambda - a.e$$

Thus $\infty \in P_2$

Actually $g \in \mathcal{L}^\infty(\lambda) \Rightarrow g \in \mathcal{L}^0(\lambda)$

Since

$$|g| \leq R \quad \lambda - a.e \Leftrightarrow \lambda(\{|g| > R\}) = 0$$

For $t \rightarrow \infty$ we will at some point have $R < t \Rightarrow \lambda(\{|g| \geq t\}) = 0$

And thus $\lim_{t \rightarrow \infty} \lambda(\{|g| \geq t\}) = 0 \Rightarrow 0 \in P_2$

Now to $p \in (0, \infty)$

Again we consider the function

$$|g|^p(x) = \begin{cases} \frac{1}{x^p} & \text{if } x \in (1, \infty) \\ 0 & \text{otherwise} \end{cases}$$

We need to consider the integral again

$$\int_{\mathbb{R}} |g|^p d\lambda = \int_{\mathbb{R}} \mathbb{I}_{(1, \infty)} \frac{1}{x^p} \lambda(dx) = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} \mathbb{I}_{[1+\frac{1}{n}, n]} \frac{1}{x^p} \lambda(dx)$$

Again we calculate the Lebesgue integral as the limit of Riemann integrals

$$= \lim_{n \rightarrow \infty} R \int_{1+\frac{1}{n}}^n \frac{1}{x^p} dx$$

Then we have a look at the integral

For $p \neq 1$

$$R \int_{1+\frac{1}{n}}^n \frac{1}{x^p} dx = \frac{1}{1-p} [x^{1-p}] = \frac{1}{1-p} \left(n^{1-p} - \left(1 + \frac{1}{n}\right)^{1-p} \right)$$

For $p < 1$

$$\frac{1}{1-p} \left(n^{1-p} - \left(1 + \frac{1}{n}\right)^{1-p} \right) \rightarrow \infty \text{ for } n \rightarrow \infty$$

Since $n^{1-p} \rightarrow \infty$ and $\left(1 + \frac{1}{n}\right)^{1-p} \rightarrow 1$

Thus

For $p > 1$

$$\frac{1}{1-p} \left(n^{1-p} - \left(1 + \frac{1}{n} \right)^{1-p} \right) \rightarrow -\frac{1}{1-p} \text{ for } n \rightarrow \infty$$

Since $n^{1-p} \rightarrow 0$, $\left(1 + \frac{1}{n} \right)^{1-p} \rightarrow 1$

For $p = 1$

$$R \int_{1+\frac{1}{n}}^n \frac{1}{x^p} dx = [\log(x)]_{1+\frac{1}{n}}^n = \log(n) - \log\left(1 + \frac{1}{n}\right) \rightarrow \infty \text{ for } n \rightarrow \infty$$

since $\log(n) \rightarrow \infty$, $\log\left(1 + \frac{1}{n}\right) \rightarrow 0$

thus we have $P_2 = (1, \infty] \cup \{0\}$